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(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR DETECTING FAKE NEWS USING MACHINE LEARNING AND DEEP LEARNING PROCESSING ARCHITECTURES A computer-implemented system for detecting fake news articles using machine learning and deep learning processing architectures is disclosed. The system comprises a processor coupled to a memory, wherein the processor executes instructions stored in the memory to implement multiple processing modules. The system includes a data collection module configured to receive news article data from user devices over a network, a text preprocessing module configured to perform tokenization, stop word removal, lemmatization, punctuation elimination, and whitespace normalization operations on the received data, and a feature extraction module configured to generate term frequency-inverse document frequency vectors, word embedding vectors, and contextual embedding vectors from the preprocessed text. The system evaluates classification outputs using accuracy, precision, recall, and F1-score metrics to enable comparative analysis of model performance and generates final classification results with confidence scores indicating whether news articles are fake or true. [To be published with Figure 1]

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